Knowledge-Driven Category Representation Learning for Remote Sensing Classification of Coastal Wetlands

Binge Cui, *Member, IEEE*, Dongrui Lv, Huan Sun, Guangbo Ren, Haojie Li, and Yan Lu

***Abstract*— Fine-grained classification of coastal wetlands from remote sensing images is a challenging task due to the spectral overlap between different wetland vegetation types, making them difficult to distinguish. Traditional methods for remote sensing interpretation often rely on manual classification or shallow machine learning approaches, which fail to effectively capture complex spatial relationships and contextual information. To integrate ecological and biological prior knowledge and enhance the generalization performance of the model, this article proposes a remote sensing image classification method for coastal wetlands based on category representation learning, called CRLNet. The core idea is to learn category-invariant representations of land-cover types in coastal wetlands using geoscience knowledge graphs (KGs) and deep neural networks. First, deep feature maps and classification probability maps generated by a semantic segmentation network are used to initialize the representations of each category; then, the spatial topological relationship embedding (STRE) and category attribute knowledge embedding (CAKE) are proposed, employing a two-stream architecture to refine the representations of each category; finally, each pixel is assigned to the category with the highest similarity based on the aforementioned deep feature maps and category-invariant representations. By combining graph convolution and self-attention mechanisms, CRLNet effectively integrates ecological and biological prior knowledge into category representation learning, thereby reducing the likelihood of conflicts between classification results and geoscience prior knowledge. The experimental results demonstrate that CRLNet outperforms state-of-the-art methods on the Huanghe River and Yancheng coastal wetland datasets. Notably, CRLNet is a lightweight framework with only 1/55th of the parameter count of CGGLNet, making it computationally efficient while maintaining high classification accuracy. The source code is available at https://github.com/cuibinge/CRLNet.**

***Index Terms*— Coastal wetland, deep learning, feature enhancement, knowledge graph (KG), remote sensing.**

Received 17 January 2025; revised 15 April 2025; accepted 14 May 2025. Date of publication 19 May 2025; date of current version 29 May 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42276185 and Grant 42076189, in part by Shandong Province Natural Science Foundation of China under Grant ZR2020MD096 and Grant ZR2020MD099, and in part by Taishan Scholar Program of Shandong Province under Grant tstp20221128. *(Corresponding author: Yan Lu.)*

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Digital Object Identifier 10.1109/TGRS.2025.3571291

I. Introduction

COASTAL wetlands, situated between terrestrial and marine ecosystems, are essential for maintaining environmental stability, preserving biodiversity, and providing renewable resources. Often referred to as "biological gene banks," these ecosystems play a crucial role in influencing the Earth's ecological balance [1], [2]. However, global climate change and human activities have increasingly degraded their ecological integrity, threatening their functions [3], [4]. Safeguarding these wetlands has become a critical priority, with remote sensing interpretation emerging as an indispensable tool to assess and quantify ecosystem services [5].

Traditional coastal wetland remote sensing interpretation methods have several limitations. Manual interpretation, though widely adopted, is time-consuming and demands significant domain expertise [6]. Machine learning approaches such as support vector machines, decision trees, and random forests primarily rely on shallow features [7], [8]. These methods struggle to capture the complex spatial patterns and contextual relationships inherent in land-cover data. As a result, their classification performance in coastal wetland environments is often suboptimal.

The advent of deep learning has significantly advanced remote sensing interpretation [9]. Models such as convolutional neural networks (CNNs), fully convolutional networks (FCNs), and transformers have been widely employed in wetland classification tasks [10], [11]. Hosseny et al. [12] proposed a spatiotemporal ensemble deep learning model for wetland classification, while Jamali et al. [13], [14] utilized the Swin transformer to classify wetlands using Sentinel-1 and Sentinel-2 data. Pan [15] proposed a 3-D convolution method based on feature sequences for wetland classification from multispectral images. Han et al. [16] proposed a CNN-based feature intersecting learning (FIL-NN) for wetland classification using multisource remote sensing data, which exhibits better performance in wetland scenes. Liu et al. [17] proposed a multistream CNN based on the fusion of hyperspectral imagery (HSI) and multispectral imagery (MSI) from GF-5 and Sentinel-2 data for coastal wetland information extraction. You et al. [18] applied the VGG16 network to classify features in Yancheng coastal wetland. Cui et al. [19] proposed MFVNet, a coastal wetland information extraction network combining deep learning and vegetation index, demonstrating extraction accuracy in the Huanghe River Estuary superior to

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traditional machine learning classification methods. Feng et al. [20] proposed a multibranch deep CNN (DCNN) that employs multitemporal data from different sensors to extract information from Sentinel images in the Huanghe River Estuary wetlands. Recent studies have further expanded the application of multimodal collaboration [21] and self-supervised learning techniques [22]. Lin et al. [23] proposed SS-MAE, enhancing spatial–spectral feature extraction in multisource classification via joint reconstruction of spatial patches and spectral channels, integrating global–local features. Gao et al. [24] developed MSFMamba, enhancing long-range dependence and interaction in HSI-LiDAR and SAR fusion with multiscale Mamba modules and dual-input fusion. Wang et al. [25] developed NNCNet, addressing heterogeneous gaps and semantic misalignment via nearest-neighbor augmentation and bilinear attention, enabling high-order HSI-LiDAR interactions for discriminative joint classification. Liu et al. [26], [27] developed a method for multispectral–hyperspectral collaborative classification through a class-guided coupled dictionary learning approach and separable coupled dictionary learning. Dong et al. [28] introduced a generative CNN foundation model, integrating sparse modeling with a low-frequency reconstruction framework, to extract generalizable features via self-supervised learning. In addition, generative adversarial techniques [29] have provided new perspectives for interpreting complex scenes. Ren et al. [30] designed the Confucius trilearning paradigm, which jointly trains dual classifiers and a generator, leveraging learning from both "good" and "bad" samples to enhance classification performance under limited annotation conditions. However, these approaches predominantly focus on data features and network structures while neglecting the use of geographic knowledge. In complex environments like coastal wetlands, where spectral and textural similarities between categories (e.g., reeds and *Spartina alterniflora*) are common, this limitation leads to classification challenges [31].

Recent research has underscored the potential of integrating knowledge reasoning with data learning for remote sensing interpretation [32], [33]. Ni et al. [34] developed a framework that integrates object counting into image captioning, thereby improving the informativeness and accuracy of remote sensing image descriptions. Dong et al. [35] proposed a unified parameter-efficient fine-tuning (UPetu) framework, which enhances the transferability of foundational remote sensing models to dense tasks through the use of quantized adapters and context-aware prompt modules. Li et al. [36] proposed a collaboratively boosting framework (CBF) that iteratively combines a data-driven deep learning module with a knowledge-guided ontology reasoning module, thereby enhancing the interpretability and reliability of deep learning-based remote sensing image interpretation methods. Yeo et al. [37] utilized existing field observations as prior knowledge, coupled with high-spatial-resolution multispectral images containing a limited number of bands and radiometric indices, to perform vegetation classification and mapping of salt marsh habitats. Similarly, Li et al. [38] treated global land-cover products and geographic locations associated with remote sensing images as geographic knowledge, offering supervised information for representation learning and network pretraining. Cui et al. [39] introduced a knowledge- and geographic object-based graph convolutional network (KGGCN), which models hyperpixel blocks as nodes within a graph network and integrates prior knowledge with spatial relevance during the information aggregation process. Furthermore, techniques leveraging knowledge graphs (KGs) to incorporate spatial relationships and semantic reasoning have shown significant promise in improving classification accuracy. However, despite these advancements, the application of KGs in combination with deep learning for coastal wetland remote sensing remains largely underexplored.

This article proposes a novel method for coastal wetland classification that combines geographic knowledge with deep learning. The key contributions are as follows.

1) *KG Construction:* A coastal wetland KG is developed, incorporating spatial relationships and attribute knowledge (e.g., plant taxonomy, shape, and texture) to improve category differentiation in remote sensing imagery.
2) *Novel Representation Learning Modules:* Two modules, the spatial topological relationship embedding (STRE) and the category attribute knowledge embedding (CAKE), are introduced to encode spatial and attribute knowledge, enabling the model to handle complex and ambiguous classification scenarios.
3) *Integration of Knowledge and Data-Driven Learning:* A coupling framework is designed to integrate geoscientific knowledge with deep feature extraction, enhancing the model's generalization and interpretability.

II. Proposed Method

The coastal wetland remote sensing interpretation method is shown in Fig. 1. First, a geoscience KG is constructed for coastal wetland classification. A lightweight global–local semantic enhancement (GLSE) module extracts features from the input image to generate semantic feature maps. These maps are also passed through a network of classification heads to generate probability maps for each category. The semantic feature maps are then multiplied by the category probability maps to derive preliminary representation vectors for each category. We refine these vectors by incorporating spatial topology and attribute knowledge from the geoscience KG. Finally, we compute the correlation between each pixel's feature vector and the category representation vectors. The category with the highest correlation is assigned to each pixel.

A. Construction of KG for Coastal Wetlands

The data sources for constructing the coastal wetland geoscience KG include Baidu Baike,[1] the Database of Coastal Bioresources of China,[2] and the Ecological Environment and Biodiversity Information System in the Yellow River Delta,[3] among others. These heterogeneous sources often

[1] https://baike.baidu.com
[2] http://plant.yic.ac.cn
[3] http://bhyrd.yic.ac.cn/home/main/index

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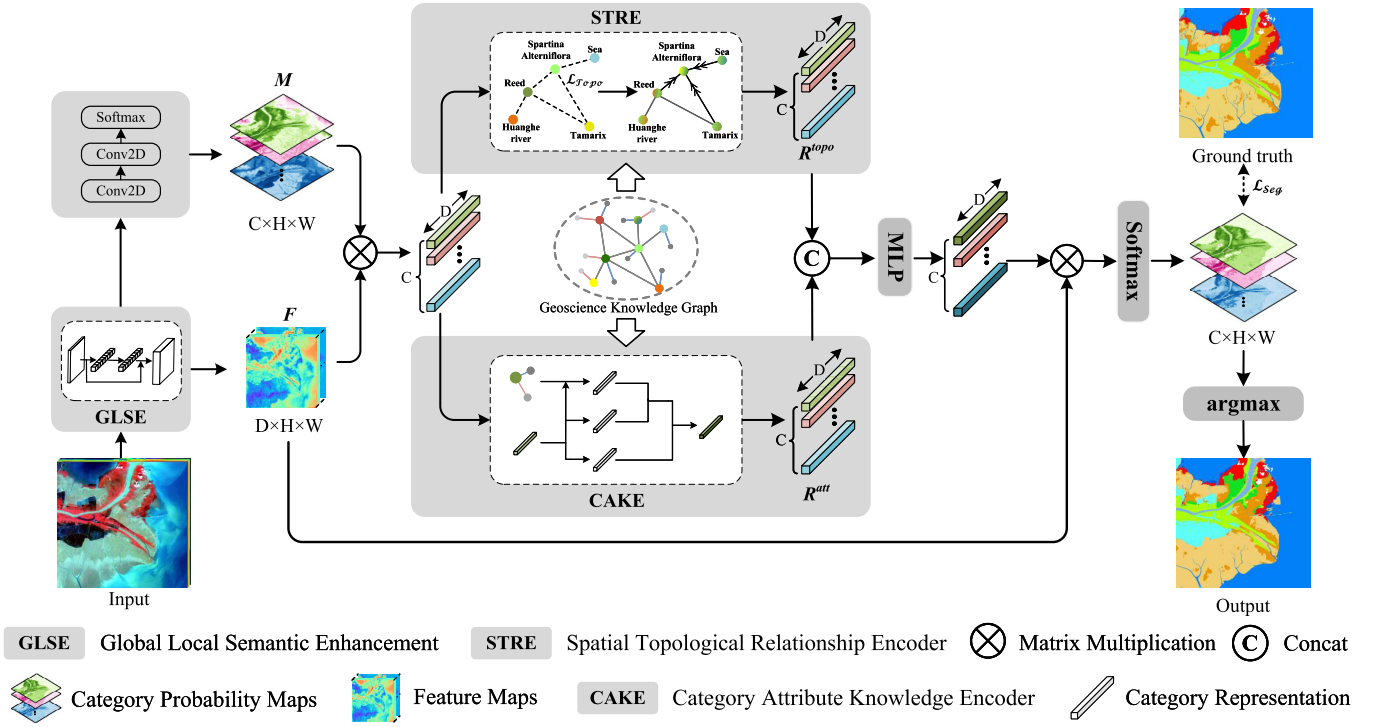


Fig. 1. Schematic illustration of the CRLNet. Initially, the input image undergoes feature extraction to obtain semantic features. Subsequently, coarse segmentation methods are employed to process these features, generating probability maps. These probability maps then serve as masks for extracting embeddings for each probability map. Based on the embeddings of these probability maps, we retrieve relevant subgraphs from the KG. Then, spatial topological neighborhood relationship and attribute knowledge of the geo-elements are encoded into the semantic representation of the geo-element categories. Finally, the enhanced features are fused to derive the ultimate outcome.

contain noise, redundancy, and inconsistencies. To ensure data quality, a multistep preprocessing pipeline is applied. First, we remove garbled text, duplicates, and irrelevant elements such as watermarks, page numbers, and extraneous symbols. Second, we correct obvious errors and fill in missing data where possible. When conflicting information is encountered, we refer to the most recent and authoritative literature to verify and resolve discrepancies. By cross-checking original sources and comparing multiple entries, we select the most reliable information, ensuring that records related to plants, communities, water bodies, tidal flats, and wetlands are accurate and nonredundant. Additionally, species names, units of measurement, and other formats are standardized to enhance consistency and data interoperability.

In the process of KG construction, the entity types, attributes, and relation types are first defined. The entity types primarily consist of plants (e.g., reeds, *Suaeda salsa*, tamarisk, *S. alterniflora*, and seagrass beds), water bodies (e.g., sea, ponds, rivers, tidal creeks, and aquaculture areas), and other land-cover types (e.g., tidal flats, farmland, and roads). Entity attributes include properties such as color, shape, elevation, slope, density, texture, lifespan, and taxonomy. Relation types cover spatial topological relationships (e.g., touches, overlap, and connectivity) and taxonomic hierarchies (e.g., family, genus, and species). Next, the KG generator (KGG) is developed to extract knowledge triples from textual data. Inspired by Graph Maker, KGG leverages the Llama 3 API, a state-of-the-art large language model (LLM), to automatically generate triples in the form of (entity, relation, and entity) and (entity, attribute, and value), based on predefined entity, attribute, and relation types. By processing a collection of texts, KGG produces subgraphs that are integrated into a comprehensive KG. Finally, the KG is reviewed by domain experts and stored in a Neo4j database for further use. The coastal wetland geoscience KG we developed includes over 3400 entities, more than 20 types of relationships, and over 14 000 triples. For the purpose of the experiments, a subset of this KG was extracted based on the main land-cover types in the study area. This subset consists of 101 triples, which feature 12 entities, including reed and *S. alterniflora*, spatial relationships such as "touches" and "disjoint," and attributes such as color and shape.

B. Category Representation Learning

1) Global–Local Semantic Enhancement: The GLSE module shown in Fig. 2 is designed to enhance feature maps by integrating global semantic context with fine-grained local details. This module achieves high-quality feature extraction with reduced computational complexity and model size, making it suitable for high-resolution image analysis tasks like remote sensing.

To extract global features, the module applies a 4×4 convolution with a stride of 4 to downsample the input, producing 64 global embedding vectors. These embeddings are refined using a multihead self-attention mechanism to capture relationships between them. The optimized embeddings are reshaped into feature maps, smoothed with a 3×3 convolution, and upsampled back to the original resolution using bilinear

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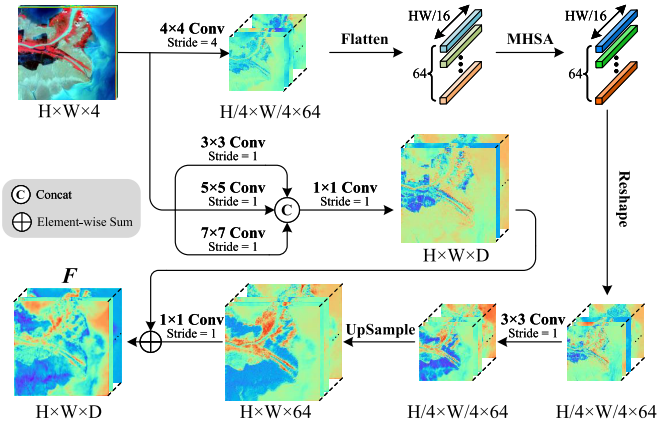


Fig. 2. Illustration of the GLSE.

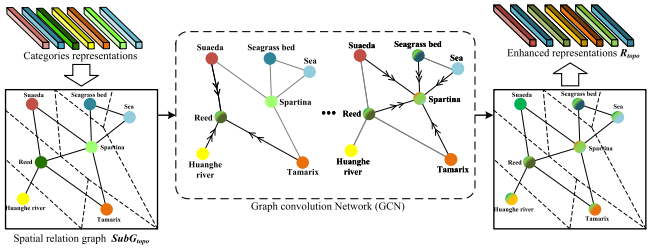


Fig. 3. Illustration of the STRE.

interpolation, followed by a 1×1 convolution to match the desired channel dimensions.

For local detail extraction, the module uses multiscale convolutions with kernel sizes of 3×3, 5×5, and 7×7 to capture features at different spatial scales. These outputs are unified through a 1×1 convolution and then fused with global semantic features to create comprehensive semantic feature maps $\boldsymbol{F}$.

By combining global and local information, the GLSE module provides a balanced representation of both context and detail, enabling precise and efficient feature extraction.

2) Category-Specific Embedding Extraction via Coarse Segmentation: The semantic feature maps $\boldsymbol{F}$ are then passed through a coarse segmentation module, consisting of two 2-D convolutional layers and a softmax activation, to generate probability maps $\boldsymbol{M} = \{m_1, m_2, \ldots, m_C\}$, where $m_i \in \mathbb{R}^{H \times W}$ represents the confidence score for each pixel belonging to category i.

The probability maps $\boldsymbol{M}$ serve as soft masks, which are used to compute category-specific embeddings $E = \{e_1, e_2, \ldots, e_C\} \in \mathbb{R}^{D \times C}$. This is achieved by reshaping $\boldsymbol{F}$ into a 2-D tensor of size $HW \times D$ and $\boldsymbol{M}$ into $HW \times C$, followed by matrix multiplication. Each embedding e_i is a D-dimensional vector summarizing the semantic features of the ith category.

This structured process enables effective initialization of category representations, setting the stage for subsequent refinement through knowledge-driven spatial and attribute encoding.

3) Spatial Topological Relationship Encoder: The STRE module, depicted in Fig. 3, is the core component of this method, aiming to optimize the semantic representation of geographic features and capture their spatial distribution characteristics. The STRE models spatial relationships between land-cover categories by constructing subgraphs based on touches and disjoint relationships from a KG. It generates two adjacency matrices, one for the touches relationships and another for the disjoint relationships. Initial embeddings are assigned to the nodes of these subgraphs, representing their features. To refine the embeddings, geometric constraints are applied in the feature space. For the touches subgraph, connected nodes are brought closer by minimizing the Euclidean distance between their embeddings. For the disjoint subgraph, a margin is enforced to ensure a minimum separation distance. These geometric constraints are integrated into a Topological Loss

$$L_{\text{topo}} = \lambda \sum_{(i,j)\in\mathcal{T}} \|e_i - e_j\|^2 + (1-\lambda) \sum_{(i,j)\in\mathcal{D}} \max\left(0, m - \|e_i - e_j\|^2\right) \quad (1)$$

where $\mathcal{T}$ and $\mathcal{D}$ denote the edge sets of the touches and disjoint subgraphs, respectively, e_i and e_j are the category-specific embeddings of categories i and j, respectively, m is the predefined geometric margin, and λ is a learnable parameter balancing the two terms.

After imposing geometric constraints, a graph convolutional network (GCN) is applied to the spatial relation subgraph $\text{SubG}_{\text{topo}}$ to update node embeddings iteratively. For the touches and disjoint subgraphs, the embeddings at layer $l+1$ are updated as follows:

$$H^{(l+1)} = \sigma(A \cdot H^{(l)} \cdot W^{(l)}) \quad (2)$$

where A is the normalized adjacency matrix (A_{touches} or A_{disjoint}), $H^{(l)}$ is the embedding at layer l, $W^{(l)}$ is the weight matrix ($W^{(l)}_{\text{touches}}$ or $W^{(l)}_{\text{disjoint}}$), and σ represents the ReLU activation function.

After multiple GCN layers, the final output embeddings from the disjoint subgraph, $H^{(L)}_{\text{disjoint}}$, encode the spatial topological relationships of geographic features, providing enhanced semantic representations of land-cover categories.

By integrating spatial topological relationship modeling, geometric constraints, and graph convolution, the STRE module effectively captures the spatial distribution and relationships of geographic features, ensuring robust and accurate land-cover classification.

4) Category Attribute Knowledge Encoder: The CAKE module shown in Fig. 4 is designed to optimize category representations by integrating visual features with attribute information derived from a geoscience KG. Its primary objective is to ensure that category embeddings are aligned with attribute semantics, resulting in semantically consistent and more discriminative representations in the attribute space. The module achieves this by encoding attribute knowledge extracted from subgraphs of the KG using a pretrained BERT model [40], fusing category embeddings and attribute knowledge embeddings through a gating mechanism, and further

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refining the category representations using a query–key–value (QKV) attention mechanism.

To obtain attribute knowledge embeddings, the CAKE module begins by extracting the attribute relation subgraph SubG_{att} from the KG based on the categories represented by the embeddings. These triples encode land-cover attribute information and are converted into natural language descriptions using the template: "{Entity}'s {Attribute} is {Value}." The natural language descriptions are encoded using a pretrained BERT model to generate high-dimensional embeddings for each triple. The BERT encoder maps the textual representation into a dense vector space by applying self-attention over contextual token embeddings. This process can be expressed mathematically as follows:

$$E_{\text{BERT}} = \text{SelfAttention}(W_{\text{fc}} \cdot \text{BERT}(\text{Template}(\text{SubG}_{\text{att}})) + b_{\text{fc}}) \quad (3)$$

where Template() denotes the textual template, W_{fc} and b_{fc} are the learnable parameters of the fully connected layer, and SelfAttention applies attention to the embeddings to integrate multiple attribute representations of the same category.

To reduce the modality gap between the category-specific embeddings E from the abundance map visual features and the textual KG features E_{BERT}, the CAKE module employs a gating mechanism to dynamically balance their contributions. Specifically, by concatenating the visual representation E and the attribute representation E_{BERT}, a gating vector θ is computed as follows:

$$\theta = \sigma(W_g \cdot [E; E_{\text{BERT}}]) \quad (4)$$

where W_g is a learnable weight matrix, $[E; E_{\text{BERT}}]$ denotes the concatenation of the two feature sets, and σ is the sigmoid activation function. The fused feature A_{fused} is then computed by dynamically weighting the visual and attribute representations

$$E_{\text{fused}} = \theta \odot E + (1 - \theta) \odot E_{\text{BERT}} \quad (5)$$

where $\odot$ represents the elementwise multiplication. This gating mechanism enables the model to dynamically adjust the contribution of visual and attribute semantics for each category, depending on the context and task requirements.

To further refine the fused category representations, the CAKE module employs a QKV attention mechanism. In this process, the fused feature E_{fused} is used as the query, while the visual feature E generates both the key and the value. The QKV mechanism calculates attention weights by evaluating the relevance between the query and key features. These attention weights are then used to reweight the value feature, allowing the model to emphasize the most relevant aspects of the category representation. Mathematically, this mechanism ensures that the refined representation incorporates both visual and attribute semantics in a manner that is semantically aligned and discriminative.

The CAKE module generates an enhanced category embedding, denoted as $\boldsymbol{R}_{\text{att}}$, which combines visual features and attribute knowledge. This embedding is semantically aligned and refined through the QKV attention mechanism. This final representation is well-suited for downstream tasks, as it effectively balances visual information and attribute knowledge, resulting in highly discriminative embeddings for each category.

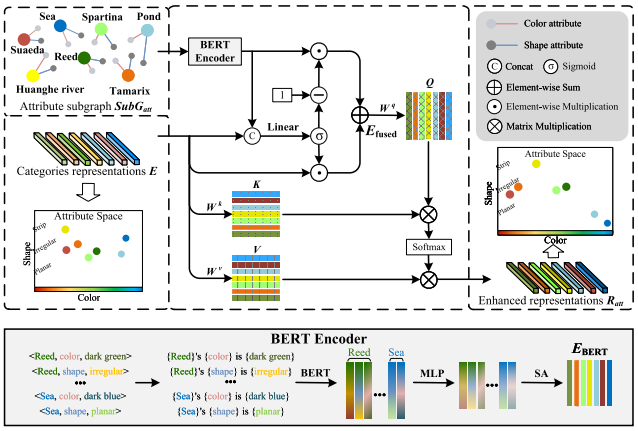


Fig. 4. Illustration of the CAKE.

C. Fine Classification of Coastal Wetlands Based on Category Representation

The optimized land-cover features $\boldsymbol{R}_{\text{topo}}$ and $\boldsymbol{R}_{\text{att}}$ are fused for the final classification. This fusion enables to fully utilize spatial information, land-cover attributes, and semantic features of images, and comprehensively consider multimodal information. The concatenated features are reduced to the same dimension of $C \times HW$ by fully convolution, and the semantic feature $\boldsymbol{F}$ is reshaped into a matrix of the size of $HW \times D$ by reshape operation. This matrix is then multiplied with the concatenated features and fed to the classifier, which mainly consists of two convolution layers with 1×1 kernel size and a softmax activation function. The processes of our method are detailed in Algorithm 1.

D. Loss Function Design

To optimize classification performance while integrating geoscientific prior knowledge, a composite loss function is designed by combining Focal Loss as the segmentation loss and a weighted Topological Loss. This design ensures the model captures both semantic and spatial relationships within coastal wetland land cover effectively.

Focal Loss is used to address class imbalance issues in pixelwise classification tasks by emphasizing hard-to-classify regions. It dynamically adjusts the contribution of well-classified pixels, enhancing the model's focus on underrepresented categories and ambiguous boundaries.

The Topological Loss L_{topo} leverages spatial relationships between land-cover categories defined by geoscientific priors, such as touches or disjoint constraints, to improve spatial semantic consistency in predictions. A dynamic weight λ_{topo} is introduced to balance the contributions of the Topological Loss during training

$$\lambda_{\text{topo}} = \frac{\lambda_{\text{init}}}{\text{epoch} + 1}. \quad (6)$$

This weighting mechanism prioritizes Topological Loss during early training stages, encouraging the model to

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Algorithm 1 CRLNet

Input: Remote sensing image $\boldsymbol{I}$, coastal wetland knowledge graph KG.

Output: Interpretation result $\boldsymbol{O}$.

1: Apply feature extraction on $\boldsymbol{I}$ using the GLSE module to obtain semantic feature maps $\boldsymbol{F}$, and use the coarse segmentation function to generate probability maps $\boldsymbol{M}$.

2: Compute category-specific embeddings $\boldsymbol{E}$ by multiplying $\boldsymbol{F}$ and $\boldsymbol{M}$.

3: Extract spatial relationship subgraph $SubG_{topo}$ and attribute subgraph $SubG_{att}$ from KG based on category embeddings $\boldsymbol{E}$.

4: **STRE module:**

   1) Assign $\boldsymbol{E}$ to the nodes of $SubG_{topo}$.

   2) Refine embeddings by applying geometric constraints (Eq. (1)) and update node embeddings using the GCN (Eq. (2)), resulting in spatial topological knowledge-enhanced land-cover features $\boldsymbol{R}_{topo}$.

5: **CAKE module:**

   1) Encode attribute subgraph $SubG_{att}$ into textual embeddings using BERT and compute E_{BERT} (Eq. (3)).

   2) Fuse visual and attribute features using the gating mechanism (Eq. (4) and Eq. (5)) and refine the representation using the QKV attention mechanism, resulting in attribute knowledge-enhanced land cover features $\boldsymbol{R}_{att}$.

6: Fuse $\boldsymbol{R}_{topo}$, $\boldsymbol{R}_{att}$ with $\boldsymbol{F}$ to obtain fused features $\boldsymbol{H}$.

7: Use a classifier on $\boldsymbol{H}$ to get the interpretation result $\boldsymbol{O}$.

8: Output $\boldsymbol{O}$.

learn spatial relationships. As training progresses, the weight decreases, allowing the model to focus more on refining semantic segmentation accuracy.

By integrating Focal Loss and the weighted Topological Loss, the model achieves comprehensive optimization, addressing challenges like land-cover confusion and complex spatial distributions in coastal wetland environments.

III. Experiment

In this article, experiments were conducted using both GF-2 remote sensing images and Sentinel-2 remote sensing images to validate the effectiveness of the proposed method. As shown in Fig. 5, the Huanghe River Estuary coastal wetland in Shandong Province and Yancheng coastal wetland in Jiangsu Province were selected as the study areas.

A. Experimental Dataset

The image of the Huanghe River Estuary coastal wetland shown in Fig. 6(a) was captured by the GF-2 satellite in August 2016 with a spatial resolution of 4 m. The GF-2 satellite is China's first domestically developed civil optical remote sensing satellite. It is equipped with two high-resolution cameras, one with 1-m panchromatic capability and the other with 4-m multispectral capability. The image of the study area is 7300 × 6908 pixels in size, encompassing four bands: red, green, blue, and near-infrared. Preprocessing, including image registration, atmospheric correction, and radiometric calibration, was performed. The study area is located within the Huanghe River Estuary National Nature Reserve in Dongying City, Shandong Province. Its approximate coordinates range from 37°35′ to 37°55′N and 119°01′ to 119°20′E. Located at the confluence of the Huanghe River and the Bohai Sea, this region is endowed with abundant water resources, rich vegetation, and unique hydrological conditions. The ionic interaction between saltwater and freshwater promotes sediment coagulation and settlement, forming expansive wetlands. Based on on-site surveys of the Huanghe River Estuary coastal wetlands, image characteristics, and expert visual interpretation, the identified land-cover categories include seagrass beds, reed, *S. alterniflora*, Tamarix mixed area, sparse vegetation, and tidal flats, among others.

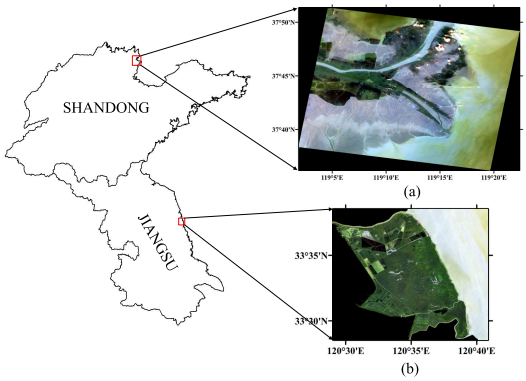


Fig. 5. Location of the study areas and experimental image. (a) Huanghe river estuary coastal wetland. (b) Yancheng coastal wetland.

The image of the Yancheng coastal wetlands shown in Fig. 6(c) was captured by the Sentinel-2 satellite. The Sentinel-2 satellite was launched by the European Space Agency in September 2020 and has a spatial resolution of 10 m. Each Sentinel-2 satellite is equipped with a multispectral imager capable of capturing images in 13 bands encompassing visible, near-infrared, and shortwave infrared. The Sentinel series offers data products such as Level-1C and Level-2A. In this article, Level-2A data are used, which has undergone data preprocessing such as geometric correction and atmospheric correction. The image of the study area is 1930 × 1860 pixels in size, and four specific bands (red, green, blue, and near-infrared) were used. The Yancheng coastal wetland is the largest and ecologically most ecologically intact mudflat coastal wetland on the Pacific coast. Due to its unique geographical environment, this wetland has become the primary center for the growth of wetland vegetation, including reed and salt-tolerant plants. The study area, located between 33°28′–33°38′N and 120°30′–120°40′E, is in the northeastern part of Yancheng City, Jiangsu Province, near the Yancheng

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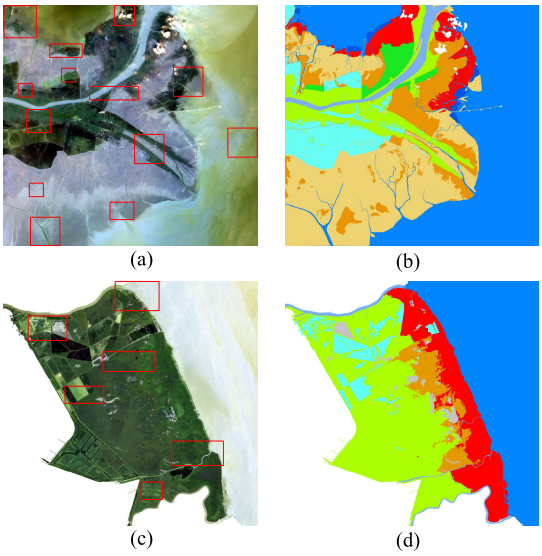


Fig. 6. Study area (the red-box region is the training area) and label image. (a) Huanghe River Estuary coastal wetland. (b) Labeled image of the Huanghe River Estuary coastal wetland. (c) Yancheng coastal wetland. (d) Labeled image of the Yancheng coastal wetland.

National Rare Bird Nature Reserve. The identified land-cover categories of the Yancheng coastal wetland include reed, *S. alterniflora*, *S. salsa*, pond, sea, and others.

In order to improve the stability of the model training and to expedite the model convergence, the z-score normalization method in the following equation was implemented:

$$B_i^{\text{normal}} = \frac{B_i - B_i^{\text{mean}}}{B_i^{\text{std}}}, \quad i = 1,2,3,4 \tag{7}$$

where B_i is the ith channel of feature map B, B_i^{mean} is the mean of B_i and B_i^{std} is the standard deviation of B_i.

B. Experimental Parameters and Evaluation Metrics

The PyTorch deep learning framework is chosen to implement the model proposed in this article, and 200 epochs of model training were conducted on NVIDIA GeForce RTX 2080Ti. The optimizer is Adam's algorithm, the learning rate is set to 0.0001, and the training batch size is set to 8. We selected 20% of the image as training samples [red-box region in Fig. 5(a)] and the rest as test samples. For each category, precision (P), recall (R), and $F1$-score ($F1$) were used as metrics to evaluate the interpretation results. These metrics are based on the confusion matrix, which contains four items: true positive (TP), false positive (FP), true negative (TN), and false negative (FN). Pixel-by-pixel comparison was used to compare the decoded results with the ground truth. The formulas for precision, recall, and $F1$-score are as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{9}$$

$$F1 = \frac{2 \times P \times R}{P + R}. \tag{10}$$

Overall accuracy (OA) and Kappa coefficient were adopted as the overall evaluation metrics for the classification results. The calculation of the Kappa coefficient is presented in the following equations:

$$\text{OA} = \frac{n_1 + n_2 + \cdots + n_C}{n} \tag{11}$$

$$P_e = \frac{t_1 \times p_1 + t_2 \times p_2 + \cdots + t_C \times p_C}{n \times n} \tag{12}$$

$$\text{Kappa} = \frac{\text{OA} - P_e}{1 - P_e} \tag{13}$$

where n is the total number of samples, C is the number of categories, n_i $(1 \le i \le C)$ represents the number of samples correctly classified as category i, t_i represents the number of true samples of category i, and p_i represents the number of samples predicted as category i.

C. Experimental Results

To verify the validity of the model proposed in this article, comparison experiments with several related works were conducted on both the Huanghe River Estuary coastal wetland and Yancheng coastal wetland datasets. Comparison methods include the latest deep neural network-based semantic segmentation methods such as CTCFNet [41], CMTFNet [42], SRCBTFusion-Net [43], and CGGLNet [44], lightweight semantic segmentation methods such as LW-IRSTNet [45] and WetMapFormer [46], and knowledge-driven methods such as LDGNet [47] and KGGCN [39].

1) Comparative Experiments on Huanghe River Estuary Coastal Wetland Dataset: Fig. 7(b)–(e) shows the classification results of deep neural network-based semantic segmentation methods, which can capture rich contextual information and multiscale features and produce relatively accurate classification results. However, they all show some degrees of category confusion, as shown in the red boxes. CTCFNet has insufficient ability to distinguish between *S. alterniflora* and reed; CMTF misclassifies sea near the abandoned Huanghe River Estuary as Huanghe river, and there are obvious omissions of *S. salsa*; SRCBTFusion-Net has the confusion of *S. alterniflora* and Tamarix mixed area, and part of the pond is also misclassified as sea; CGGLNet has poor recognition of seagrass bed in the coastal area, with a lot of omissions. As shown in Fig. 7(f) and (g), lightweight segmentation methods experience more category confusion. For example, large areas of pond are misclassified as sea, and WetMapFormer performs poorly at identifying sea in tidal ditches. Although LDGNet enhances visual feature extraction through semantic descriptions, and KGGCN integrates prior knowledge with spatial correlations for improved information aggregation, both methods still suffer from notable confusion between *S. alterniflora* and reed. In contrast, our CRLNet fully incorporates the spatial topology and attribute prior knowledge of land-cover types, which enables more accurate remote sensing classification in complex coastal wetland environments. For example, sea and pond are not adjacent, while *S. alterniflora* is adjacent to the sea but far from pond. Attribute knowledge also enhances the ability to distinguish between different land covers, effectively mitigating the problem of category confusion.

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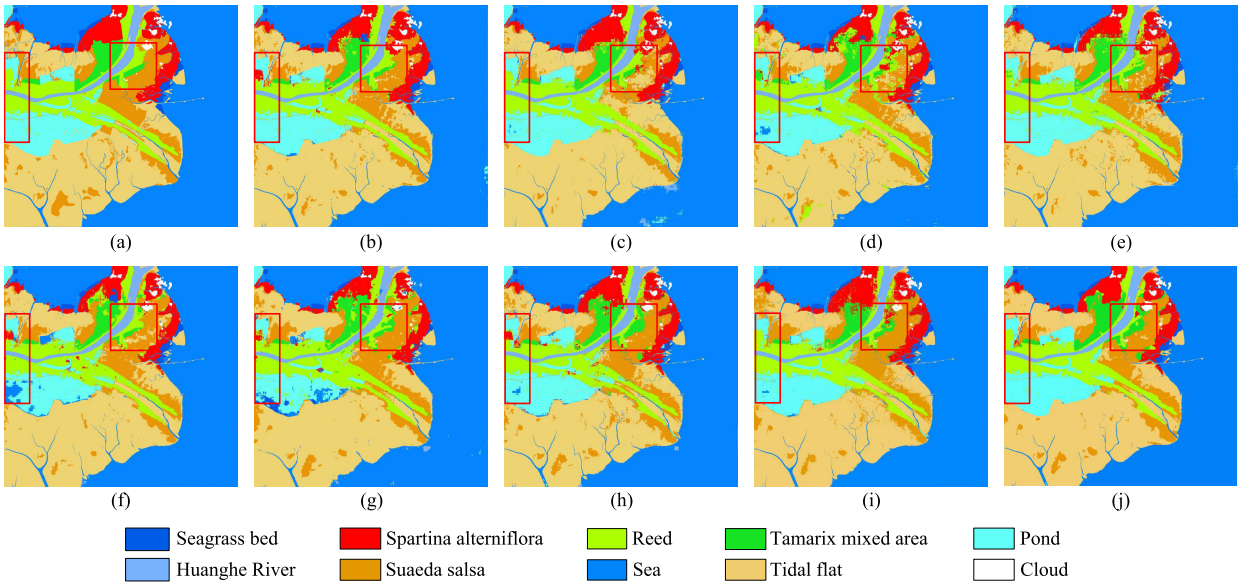


Fig. 7. Interpretation results of different methods on the Huanghe River Estuary coastal wetland dataset. (a) GT. (b) CTCFNet. (c) CMTFNet. (d) SRCBTFusion-Net. (e) CGGLNet. (f) LW-IRSTNet. (g) WetMapFormer. (h) LDGNet. (i) KGGCN. (j) Our CRLNet. Among them, (b)–(e) are the latest semantic segmentation methods based on deep neural networks, (f) and (g) are lightweight semantic segmentation methods, and (h) and (i) are knowledge-driven methods.

TABLE I

ACCURACY COMPARISON ON THE HUANGHE ESTUARY COASTAL WETLAND DATASET

Class	CTCFNet [41]			CMTFNet [42]			SRCBTFusion-Net [43]			CGGLNet [44]			LW-IRSTNet [45]			WetMapFormer [46]			LDGNet [47]			KGGCN [39]			CRLNet(Ours)		
	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1
Seagrass bed	96.1	57.7	0.721	95.6	77.4	**0.855**	91.7	74.7	0.823	**97.3**	66.5	0.790	68.7	**79.5**	0.737	51.9	50.2	0.510	80.5	63.6	0.710	77.7	69.8	0.735	94.9	77.4	0.852
Spartina alterniflora	82.4	90.7	0.863	87.5	**94.9**	**0.910**	82.5	87.4	0.849	84.9	90.9	0.878	88.3	77.8	0.827	81.2	74.7	0.778	81.8	87.7	0.846	88.6	86.8	0.877	**90.5**	84.5	0.873
Reed	92.7	92.5	**0.926**	87.4	**94.3**	0.907	89.3	86.5	0.879	89.2	92.9	0.910	83.5	92.4	0.877	80.0	91.2	0.852	**93.4**	85.2	0.891	90.1	91.7	0.909	90.9	**94.3**	**0.926**
Tamarix mixed area	78.0	**89.9**	0.835	**91.3**	83.1	**0.870**	71.2	88.7	0.790	89.4	80.4	0.847	81.1	73.5	0.771	64.7	77.5	0.705	69.1	84.5	0.760	79.4	80.9	0.801	78.3	82.5	0.803
Tidal flat	91.2	**96.2**	0.936	87.2	95.3	0.911	90.9	94.2	0.925	90.5	94.4	0.924	92.0	94.5	0.932	89.9	94.4	0.921	91.7	95.0	0.933	90.7	95.1	0.928	**92.9**	94.7	**0.938**
Suaeda salsa	**87.5**	74.4	0.804	84.0	58.9	0.693	81.9	71.4	0.763	81.1	72.7	0.767	82.5	79.2	0.808	81.4	77.3	0.793	84.0	77.3	0.805	83.5	74.6	0.788	84.3	**81.3**	**0.828**
Sea	96.1	94.6	**0.954**	93.1	**94.7**	0.939	95.0	93.3	0.941	**96.3**	94.5	**0.954**	**96.3**	88.0	0.920	96.1	72.6	0.827	93.7	91.6	0.927	94.9	87.2	0.909	96.0	94.6	0.953
Huanghe River	91.0	97.7	0.942	96.5	97.2	0.968	95.4	96.8	0.961	**97.1**	97.6	**0.974**	92.4	96.8	0.945	92.0	90.1	0.911	94.6	**97.8**	0.962	96.0	96.1	0.961	96.9	97.2	0.971
Pond	99.4	98.3	0.988	99.2	98.3	0.988	98.8	98.7	0.988	**99.5**	98.9	**0.992**	98.2	98.0	0.981	97.1	97.1	0.971	98.8	98.8	0.988	97.9	98.7	0.983	99.0	**99.0**	0.990
Cloud	88.4	94.3	0.913	87.4	93.7	0.905	88.7	94.0	0.913	90.0	**95.3**	**0.926**	86.0	92.0	0.889	**91.6**	66.2	0.769	87.6	92.3	0.899	81.2	89.4	0.851	89.9	93.9	0.919
OA(%)		93.7			92.4			92.5			93.3			92.4			90.0			92.9			92.7			**94.2**	
Kappa		0.918			0.900			0.901			0.912			0.900			0.868			0.907			0.904			**0.923**	

Table I demonstrates the quantitative results of the comparative experiments on the Huanghe River Estuary coastal wetland dataset. Our proposed CRLNet achieves the highest scores in both OA and Kappa coefficient, with an OA of 94.2% and a Kappa coefficient of 0.923, outperforming all the other methods. It is worth noting that CRLNet is a lightweight method with only a 1/55th of the model parameter count compared to CGGLNet, which has the second highest classification accuracy. Furthermore, CRLNet achieves the highest or second-highest $F1$-score for multiple categories. In particular, the $F1$-score for *S. salsa* is 0.828, which is 0.02 higher than the second-best LW-IRSTNet.

2) Comparative Experiments on Yancheng Coastal Wetland Dataset: Fig. 8 presents a performance comparison between CRLNet and other methods on the Yancheng wetland dataset. As shown in the red-box area of Fig. 8(b)–(e), CTCFNet and SRCBTFusion-Net show considerable confusion when classifying *S. alterniflora* and reed; CMTFNet and CGGLNet, while partially alleviating the confusion between *S. alterniflora* and reed, still exhibit obvious misclassification of pond. This indicates that optimizing visual feature extraction through a deep neural network alone is insufficient to fully solve the confusion problem. Lightweight methods perform poorly on the Yancheng wetland dataset, as we can see in the red-box area of Fig. 8(f) and (g), especially in distinguishing vegetation such as *S. alterniflora*, *S. salsa*, and reed. As shown in Fig. 8(h), LDGNet improves classification performance by introducing textual descriptions and enhancing attribute-level supervision. However, it struggles to distinguish between *S. alterniflora*, reed, and pond. KGGCN, as shown in Fig. 8(i), achieves better overall feature extraction, but suffers from misclassification within reed regions, where numerous small patches are incorrectly labeled as *S. salsa*. In addition, due to limited training samples for the Huanghe River class, both LDGNet and KGGCN perform poorly in this category. In comparison, CRLNet demonstrates superior ability to resolve such ambiguities by jointly modeling semantic attributes and spatial relationships through the CAKE and STRE modules. This leads to significantly more accurate and semantically consistent classification outputs.

Table II demonstrates the quantitative results of the comparative experiments on the Yancheng coastal wetland dataset.

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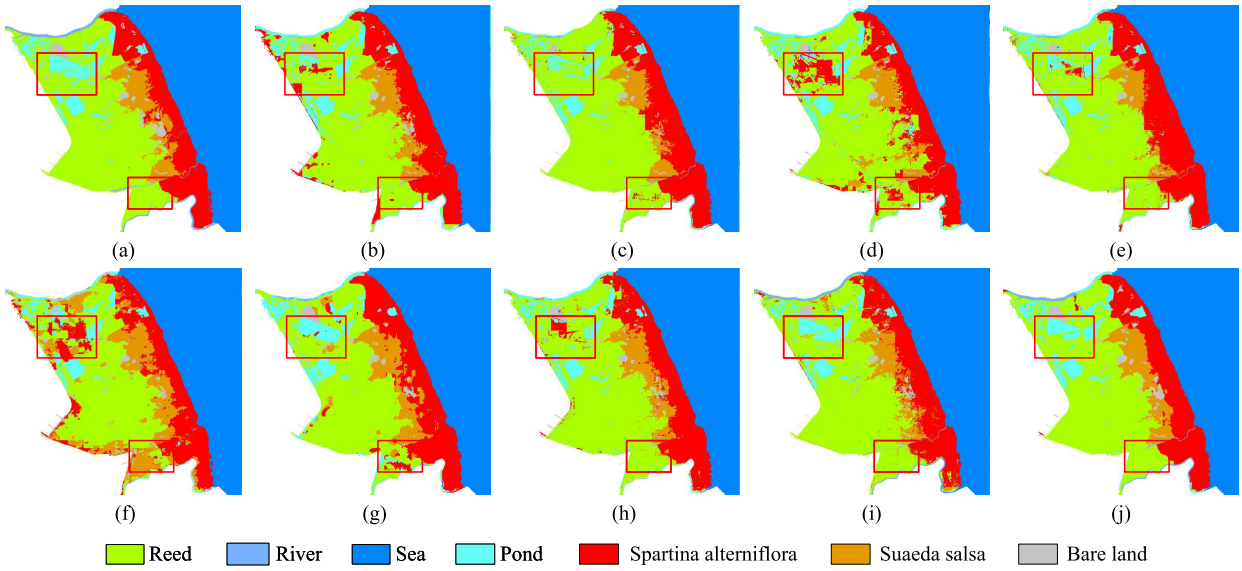


Fig. 8. Interpretation results of different methods on the Yancheng coastal wetland dataset. (a) GT. (b) CTCFNet. (c) CMTFNet. (d) SRCBTFusion-Net. (e) CGGLNet. (f) LW-IRSTNet. (g) WetMapFormer. (h) LDGNet. (i) KGGCN. (j) Our CRLNet. Among them, (b)–(e) are the latest semantic segmentation methods based on deep neural networks, (f) and (g) are lightweight semantic segmentation methods, and (h) and (i) are knowledge-driven methods.

TABLE II
ACCURACY COMPARISON ON THE YANCHENG COASTAL WETLAND DATASET

Class	CTCFNet [41]			CMTFNet [42]			SRCBTFusion-Net [43]			CGGLNet [44]			LW-IRSTNet [45]			WetMapFormer [46]			LDGNet [47]			KGGCN [39]			CRLNet(Ours)		
	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1
Spartina alterniflora	88.0	86.2	0.871	85.6	**97.3**	0.911	74.8	88.0	0.809	84.5	96.2	0.900	79.8	78.5	0.791	79.5	91.9	0.852	89.8	86.4	0.880	86.9	92.6	0.896	**93.6**	91.7	**0.927**
Suaeda salsa	78.3	78.3	0.783	**87.5**	74.1	0.802	68.9	70.9	0.699	83.8	71.1	0.769	38.1	83.2	0.522	58.0	72.9	0.646	70.8	**85.7**	0.775	76.1	69.4	0.726	82.9	80.7	**0.818**
Reed	88.3	93.7	0.909	89.8	95.2	0.925	89.0	87.1	0.880	93.0	94.2	0.936	88.1	75.4	0.812	93.0	82.9	0.877	90.3	**96.2**	0.924	92.5	94.4	0.935	**93.8**	**96.2**	**0.950**
Pond	61.7	65.1	0.634	71.3	66.5	0.688	60.6	60.7	0.606	65.5	71.9	0.685	60.7	54.0	0.572	57.1	73.3	0.642	65.9	60.6	0.632	84.2	**83.3**	**0.837**	**86.6**	73.3	0.794
River	86.1	8.3	0.152	90.4	21.8	0.351	95.1	8.8	0.161	91.4	10.7	0.191	41.9	0.1	0.003	84.6	18.8	0.308	62.2	0.3	0.006	**95.2**	64.8	**0.771**	55.4	**87.1**	0.677
Bare land	**87.7**	60.9	0.719	81.2	40.2	0.538	68.1	48.3	0.565	82.9	47.7	0.605	32.4	34.1	0.332	74.1	53.3	0.620	69.5	58.7	0.637	87.1	57.7	0.694	79.9	**83.3**	**0.816**
Sea	99.5	99.3	0.994	99.7	98.3	0.990	99.6	99.7	**0.997**	99.3	**99.9**	0.996	99.5	99.5	0.995	99.2	99.7	0.994	99.6	99.3	0.994	99.4	99.7	0.996	**99.8**	99.6	**0.997**
OA(%)	93.3			94.2			91.3			94.3			87.6			91.5			93.5			95.0			**95.9**		
Kappa	0.912			0.925			0.888			0.926			0.840			0.890			0.916			0.935			**0.947**		

TABLE III
ABLATION STUDIES FOR THE PROPOSED METHOD ON TWO DATASETS

Dataset	Method	Params (M)	FLOPs (G)	OA (%)
Huanghe River Estuary Dataset	GLSE	0.24	1.38	92.6
	GLSE+STRE	0.41	2.85	93.4
	GLSE+CAKE	1.02	2.91	93.9
	GLSE+STRE+CAKE	1.08	3.15	**94.2**
Yancheng Dataset	GLSE	0.24	1.38	93.4
	GLSE+STRE	0.41	2.85	94.5
	GLSE+CAKE	1.02	2.91	95.2
	GLSE+STRE+CAKE	1.08	3.15	**95.9**

The proposed lightweight CRLNet achieves an OA of 95.9% and a Kappa coefficient of 0.947, significantly outperforming all the other methods.

D. Ablation Experiments

To investigate the contributions of the GLSE, STRE, and CAKE modules in the model, ablation experiments were conducted on the Huanghe Estuary coastal wetland dataset and the Yancheng coastal wetland dataset. OA was used as the evaluation metric, and the experimental results are shown in Table III.

1) Effects of GLSE: The GLSE module, as the designed baseline network, aims to enhance feature maps by combining global semantic context and fine-grained local details. The experiments show that the baseline network using only the GLSE module achieves an OA of 92.6% and 93.4% on the Huanghe Estuary and Yancheng coastal wetland datasets, respectively. These results indicate that the GLSE module can achieve high-quality feature extraction with lower computational complexity and model size, making it suitable for remote sensing image analysis tasks.

2) Effects of STRE: When the STRE module was added to the baseline network, the OA increased by 0.8% on the Huanghe Estuary coastal wetland dataset and by 1.1% on the Yancheng coastal wetland dataset. When STRE was added to the baseline network with the combined CAKE module, the OA further increased by 0.3% and 0.7% for the two datasets, respectively.

These improvements benefit from the STRE module's ability to effectively integrate the spatial topology relationships between land cover in the geographic KG, helping the model learn more comprehensive geographical feature distribution information. Meanwhile, the STRE module maintains a reasonable increase in parameter count and computational complexity, with parameters rising from 0.24 to 0.41M and computational cost increasing from 1.38 to 2.85 GFLOPs.

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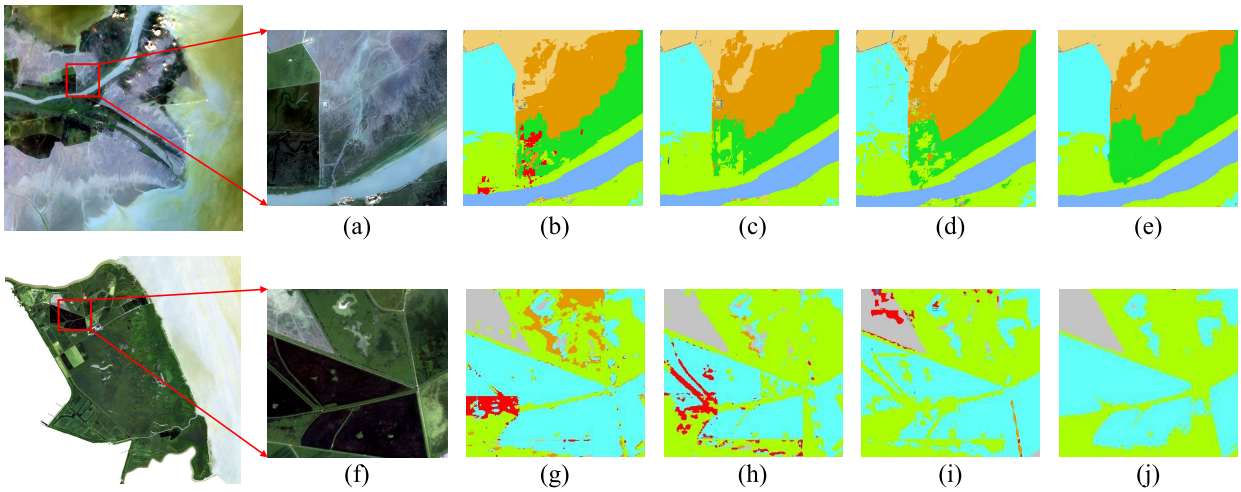


Fig. 9. Ablation experiments of local area on the Huanghe River Estuasry coastal wetland dataset and the Yancheng coastal wetland dataset. (a) and (f) Image of the local area. (b) and (g) Interpretation result of GLSE. (c) and (h) Interpretation result of GLSE + CAKE. (d) and (i) Interpretation result of GLSE + STRE, and (e) and (j) Interpretation result of GLSE + CAKE + STRE.

3) Effects of CAKE: When the CAKE module was added to the baseline network, the OA increased by 1.3% on the Huanghe Estuary coastal wetland dataset and by 1.8% on the Yancheng coastal wetland dataset. When CAKE was added to the baseline network with the combined STRE module, the OA further increased by 0.8% and 1.4% for the two datasets, respectively.

The advantage of the CAKE module lies in its effective utilization of geographic feature attribute information from the geographic KG, thereby enhancing the representation capability of geographic features and further improving classification performance. However, compared to the STRE module, the CAKE module incurs a larger increase in parameters and computational complexity, with parameters rising from 0.24 to 1.02M and computational cost increasing from 1.38 to 2.91 GFLOPs. This additional computational overhead is justified as it leads to significant improvements in accuracy.

4) Combined Effects of STRE and CAKE: When both the STRE and CAKE modules were combined with the GLSE module, the OA reached its highest levels on both datasets. Specifically, the OA on the Huanghe Estuary coastal wetland dataset reached 94.2%, and the OA on the Yancheng coastal wetland dataset reached 95.9%. These results demonstrate that the STRE and CAKE modules are highly complementary in function and can jointly enhance classification performance.

However, when both STRE and CAKE modules were introduced simultaneously, the model's parameter count and computational cost also reached their maximum, with parameters totaling 1.08M and a computational cost of 3.15 GFLOPs. Nevertheless, this computational overhead remains within an acceptable range, especially given the significant improvement in classification performance. The efficient feature extraction capability of the GLSE module, combined with the synergistic effects of STRE and CAKE, demonstrates the practicality and effectiveness of this approach for real-world remote sensing classification tasks.

Fig. 9 presents the visualization results of ablation experiments conducted on local regions of the Huanghe River Estuary coastal wetland dataset and the Yancheng coastal wetland dataset. The GLSE module provides a solid baseline by effectively extracting semantic features. Building upon this, the CAKE module models attribute characteristics to mitigate confusion between spectrally similar vegetation types, such as reeds and *S. alterniflora*. The STRE module further refines the results by introducing spatial relationship reasoning, which leverages spatial adjacency and separation to reduce misclassification between distinct land-cover categories. When combined, CAKE and STRE significantly enhance the accuracy and robustness of coastal wetland remote sensing interpretation, leading to clearer category boundaries and improved semantic consistency.

E. Effectiveness of Combining Touches and Disjoint Relations

To investigate the importance of combining touches (adjacency) and disjoint (separation) relations in the STRE for achieving optimal classification performance, we conducted experiments on the Huanghe River Estuary wetland dataset and the Yancheng wetland dataset. The experimental results are shown in Tables IV and V.

1) Limitations of Using Touches Relations Alone: When the STRE module relies solely on touches relations, the model struggles to adequately capture the global separation properties of spatial features. Although touches is advantageous in distinguishing adjacent or connected land-cover categories, such as seagrass bed and *S. alterniflora* in the Huanghe River Estuary wetland dataset, their performance is insufficient for geographically distant categories with similar spectral characteristics. For instance, in the Yancheng dataset, the OA achieved using only touches relations is merely 93.7%. This indicates that touches relations alone cannot fully capture the complexity of spatial relationships.

2) Limitations of Using Disjoint Relations Alone: On the other hand, using only disjoint relations enhances the ability to distinguish spatially separated land-cover categories, such as ponds and the sea. However, this approach overlooks

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TABLE IV

EVALUATION OF THE EFFECTIVENESS OF COMBINING TOUCHES AND DISJOINT RELATIONS IN THE STRE ON THE HUANGHE ESTUARY COASTAL WETLAND DATASET

Class	Disjoint			Touches			Disjoint + Touches		
	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1
Seagrass bed	58.4	**94.4**	0.721	81.1	92.4	**0.864**	**94.9**	77.4	0.852
Spartina alterniflora	83.1	85.3	0.842	84.2	**92.1**	**0.880**	**90.5**	84.5	0.873
Reed	**96.0**	87.3	0.914	94.0	89.3	0.916	90.9	**94.3**	**0.926**
Tamarix mixed area	80.4	81.9	**0.812**	**92.9**	71.1	0.806	78.3	**82.5**	0.803
Tidal flat	**95.3**	92.6	**0.939**	93.0	93.5	0.933	92.9	**94.7**	0.938
Suaeda salsa	79.8	**84.9**	0.823	81.7	82.1	0.819	**84.3**	81.3	**0.828**
Sea	91.7	**97.8**	0.946	92.3	96.6	0.944	**96.0**	94.6	**0.953**
Huanghe River	**97.2**	**97.6**	**0.974**	**97.2**	96.5	0.969	96.9	97.2	0.971
Pond	98.9	**99.1**	**0.990**	98.7	**99.1**	0.989	**99.0**	99.0	**0.990**
Cloud	**95.8**	86.1	0.907	92.6	88.5	0.905	89.9	**93.9**	**0.919**
OA(%)		93.8			93.7			**94.2**	
Kappa		0.918			0.918			**0.923**	

TABLE V

EVALUATION OF THE EFFECTIVENESS OF COMBINING TOUCHES AND DISJOINT RELATIONS IN THE STRE ON THE YANCHENG WETLAND DATASET

Class	Disjoint			Touches			Disjoint + Touches		
	P(%)	R(%)	F1	P(%)	R(%)	F1	P(%)	R(%)	F1
Spartina alterniflora	91.2	84.5	0.877	92.7	86.0	0.892	**93.6**	**91.7**	**0.927**
Suaeda salsa	73.4	76.1	0.747	78.5	67.0	0.723	**82.9**	**80.7**	**0.818**
Reed	**93.8**	91.8	0.928	86.2	95.8	0.908	**93.8**	**96.2**	**0.950**
Pond	77.0	**75.8**	0.764	**89.8**	68.0	0.774	86.6	73.3	**0.794**
River	28.9	87.0	0.433	**70.0**	81.0	**0.751**	55.4	**87.1**	0.677
Bare land	79.2	77.8	0.785	**83.0**	71.3	0.767	79.9	**83.3**	**0.816**
Sea	97.9	**99.8**	0.989	98.4	99.6	0.990	**99.8**	99.6	**0.997**
OA(%)		94.1			93.7			**95.9**	
Kappa		0.924			0.920			**0.947**	

critical touches information, making it difficult to differentiate categories with overlapping spectral characteristics that are spatially close to each other, such as reed and *S. alterniflora*. In the Huanghe River Estuary dataset, the OA achieved using only disjoint relations is 93.8%. While this shows an improvement in global separation, it fails to significantly enhance the model's ability to distinguish spatial features at a finer granularity. This limitation arises primarily because disjoint relations cannot effectively capture the local contextual proximity of spatially adjacent features.

3) Advantages of Combining Touches and Disjoint Relations: In STRE, the combination of touches and disjoint relations effectively compensates for the limitations of using either relation alone. By integrating adjacency and separation constraints, the model can simultaneously capture both local and global spatial interactions. This synergistic approach achieves significant improvements in classification performance: in the Huanghe River Estuary dataset, the OA reaches 94.2%, with a Kappa coefficient of 0.923; and in the Yancheng dataset, the OA increases to 95.9%, with a Kappa coefficient of 0.947. These results demonstrate that the combination of complementary spatial relations enables STRE to effectively distinguish complex land-cover categories, even in regions with ambiguous spectral characteristics. Notably, this is evident in the Yancheng dataset, where the combination reduces confusion between ponds and vegetation, and in the Huanghe River Estuary dataset, where it minimizes confusion between *S. salsa* and tidal flats.

The experimental results highlight the necessity of leveraging both touches and disjoint relations to fully exploit spatial topological information. Touches relations provide critical local context, while disjoint relations introduce essential global constraints. Together, they enable STRE to effectively model complex spatial dynamics, significantly enhancing classification accuracy and robustness.

F. Impact of Prompt and Text Encoder Choices on Attribute Knowledge Embedding Performance

To thoroughly investigate the factors influencing the performance of attribute knowledge embedding in the CAKE module, we designed an experiment to systematically evaluate the combined effects of natural language prompt structures and text encoders. Three prompt structures, such as "{entity}'s {attribute} is {value}," "The {attribute} of {entity} is {value}," and "{entity} has a {attribute} of {value}," were tested using four text encoders (BERT, DistilBERT [48], RoBERTa [49], and Sentence-BERT [50]). The evaluation was conducted on two wetland datasets (the Huanghe River Estuary wetland dataset and the Yancheng wetland dataset), using OA and the Kappa coefficient as performance metrics.

The experimental results shown in Table VI demonstrated that the choice of text encoder significantly affects the quality of attribute knowledge embeddings. Benefiting from exceptional bidirectional contextual understanding, BERT can capture subtle semantic relationships within KG triples more accurately, and consistently outperformed all the other encoders across all prompt structures and datasets. For example, when paired with the prompt "{entity}'s {attribute} is {value}," BERT achieved the highest OA of 94.2% and a Kappa coefficient of 0.923 on the Huanghe River Estuary dataset, and an OA of 95.9% and a Kappa coefficient of 0.947 on the Yancheng dataset. In contrast, the lightweight DistilBERT showed slightly inferior performance due to its simplified architecture, which constrains its capacity to capture deep semantic nuances. RoBERTa and Sentence-BERT exhibited relatively weaker performance. Although RoBERTa benefits from optimized training strategies, it failed to excel in short-text prompt scenarios. Meanwhile, Sentence-BERT, which is tailored for sentence similarity tasks, lacked the precision needed for fine-grained modeling of attribute relationships. This trend persisted in the Yancheng dataset, where BERT continued to exhibit superior performance.

Additionally, the experiment confirmed the pivotal role of prompt structures in enhancing embedding performance. Among the three tested structures, "{entity}'s {attribute} is {value}" has demonstrated the best performance in most text encoders. As previously noted, when combined with BERT, this structure achieved optimal results on both Huanghe River Estuary dataset and Yancheng dataset, outperforming both "The {attribute} of {entity} is {value}" and "{entity} has a {attribute} of {value}." This pattern held true for the other text encoders as well. The advantage of "{entity}'s {attribute} is {value}" may stem from its more natural representation

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TABLE VI

ACCURACY AND KAPPA COMPARISON OF INTERPRETATION RESULTS BASED ON DIFFERENT PROMPT STRUCTURES AND TEXT ENCODERS

Prompt	Text Encoder	Huanghe River Estuary Wetland Dataset		Yancheng Wetland Dataset	
		OA(%)	Kappa	OA(%)	Kappa
{entity}'s {attribute} is {value}.	BERT [40]	**94.2**	**0.923**	**95.9**	**0.947**
	DistilBERT [48]	93.7	0.918	94.7	0.931
	RoBERTa [49]	93.4	0.913	94.4	0.928
	Sentence-BERT [50]	93.5	0.915	94.3	0.927
The {attribute} of {entity} is {value}.	BERT [40]	**93.6**	**0.916**	**95.6**	**0.943**
	DistilBERT [48]	93.1	0.909	93.5	0.915
	RoBERTa [49]	93.1	0.910	93.2	0.913
	Sentence-BERT [50]	92.9	0.907	94.1	0.925
{entity} has a {attribute} of {value}.	BERT [40]	93.7	0.917	**95.0**	**0.936**
	DistilBERT [48]	93.2	0.911	**95.0**	0.935
	RoBERTa [49]	**93.8**	**0.919**	93.8	0.918
	Sentence-BERT [50]	93.5	0.914	93.1	0.911

TABLE VII

COMPARISON OF MODEL PERFORMANCE AND EFFICIENCY

Model	OA (%)	Params (M)	FLOPs (G)	Train. Time (min)	Inf. Mem (MiB)	Inf. Time (min)
CTCFNet [41]	93.7	78.76	117.93	210.9	1961	54.4
CMTFNet [42]	92.4	30.07	2.44	79.9	1489	45.9
SRCBTFusion-Net [43]	92.5	146.74	3.87	85.2	1468	39.6
CGGLNet [44]	93.3	55.91	11.17	84.5	1803	34.2
LW-IRSTNet [45]	92.4	**0.17**	**0.85**	**59.6**	**1177**	**33.4**
WetMapFormer [46]	90.0	0.59	1.73	78.8	1535	35.8
LDGNet [47]	92.9	35.13	1.40	96.9	2123	45.9
KGGCN [39]	92.7	0.78	7.99	83.1	1932	42.4
CRLNet	**94.2**	1.08	3.15	65.8	1725	34.6

of the direct relationship between entities and attributes, thereby enhancing semantic interpretability and enabling the text encoder to perform more precise encoding.

In summary, the experimental results highlight the critical synergy between selecting an appropriate text encoder and prompt structure for attribute knowledge embedding in the CAKE module. BERT emerged as the most effective encoder, leveraging its robust contextual modeling capabilities to generate highly discriminative and semantically consistent embeddings. Meanwhile, the prompt structure "{entity}'s {attribute} is {value}" proved to be the most effective formulation, significantly improving the module's ability to accurately represent attributes from the geoscience KG. These findings not only confirm the rationality of the CAKE module design, but also provide valuable guidance for future optimizations in prompt engineering and encoder selection within KG integration tasks.

G. Model Performance Analysis

As shown in Table VII, CRLNet achieves an excellent tradeoff between accuracy and efficiency. Training and inference times were measured on 1000 cropped images (128×128 resolution), and GPU memory usage was tested with a batch size of 1. CRLNet reaches the highest OA of 94.2%, confirming the effectiveness of its category representation learning. It has only 1.08 million parameters, far fewer than models such as CTCFNet (78.76M) and CGGLNet (55.91M), highlighting its lightweight design. While its FLOPs are slightly higher than those of the most lightweight models, this comes with a notable gain in accuracy. CRLNet's training and inference times are close to the best-performing models, and it uses significantly less GPU memory than LDGNet during inference, while still maintaining better accuracy. This demonstrates CRLNet's practicality in resource-constrained environments. In summary, CRLNet combines high accuracy with low computational cost, making it both effective and efficient for remote sensing tasks.

TABLE VIII

ACCURACY AND KAPPA COMPARISON OF INTERPRETATION RESULTS BASED ON DIFFERENT LOSS FUNCTION HYPERPARAMETERS

λ_{init}	Huanghe River Estuary Wetland Dataset		Yancheng Wetland Dataset	
	OA (%)	Kappa	OA (%)	Kappa
1	93.5	0.915	93.1	0.912
0.8	93.7	0.917	94.4	0.928
0.6	**94.2**	**0.923**	**95.9**	**0.947**
0.4	93.2	0.911	94.6	0.930
0.2	93.2	0.911	94.4	0.928
0.05	92.7	0.904	93.9	0.921

H. Selection of Loss Function Hyperparameter

The experimental results presented in Table VIII demonstrate the impact of the initial weight hyperparameter λ_{init} on the model's performance for both the Huanghe River Estuary wetland dataset and the Yancheng wetland dataset. For the Huanghe River Estuary wetland dataset, the highest OA of 94.2% and Kappa coefficient of 0.923 are achieved when $\lambda_{\text{init}} = 0.6$, indicating optimal balance between the segmentation loss and the Topological Loss. Similarly, for the Yancheng wetland dataset, the best performance is observed at $\lambda_{\text{init}} = 0.6$, with an OA of 95.9% and a Kappa coefficient of 0.947.

Notably, excessively high values of λ_{init} (e.g., 1.0) slightly reduce performance, likely due to the overemphasis on Topological Loss during early training, which may hinder the model's ability to refine pixel-level semantics. Conversely, very low values (e.g., 0.05) lead to suboptimal results, as insufficient weighting of topological constraints fails to capture spatial relationships effectively. This trend highlights the necessity of careful tuning of λ_{init} to balance semantic and spatial optimization, with $\lambda_{\text{init}} = 0.6$ emerging as the most effective value across both datasets.

I. Effectiveness Analysis of New Loss Function

To verify the effectiveness of the proposed joint loss function, we conducted ablation experiments on the Huanghe River Estuary wetland and Yancheng wetland datasets. Results are summarized in Table IX.

Using Cross-Entropy Loss alone, the model achieved 92.9% and 93.3% accuracy, with Kappa coefficients of 0.907 and 0.913 on the two datasets. These results reflect its limitations in handling class imbalance and complex spatial structures. Topological Loss led to further gains, reaching 93.6% and 94.8% accuracy, with Kappa values of 0.916 and 0.932, indicating improved spatial consistency. With Focal Loss, accuracies improved to 93.4% and 93.9%, and Kappa to 0.913 and 0.920, showing its advantage in emphasizing hard examples. The proposed joint loss function outperformed all others, achieving 94.2% accuracy and 0.923 Kappa on the Huanghe River Estuary wetland, and 95.9% and 0.947 on the Yancheng wetland. Compared with Focal Loss alone, it improved accuracy by up to 2.0% and Kappa by 0.027. Compared with Cross-Entropy Loss, the gains reached as high as 2.6% in accuracy and 0.034 in Kappa. These results confirm

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TABLE IX
ACCURACY AND KAPPA COMPARISON OF INTERPRETATION RESULTS BASED ON DIFFERENT LOSS SETTINGS

Loss	Huanghe River Estuary Wetland Dataset		Yancheng Wetland Dataset	
	OA (%)	Kappa	OA (%)	Kappa
Cross-entropy Loss	92.9	0.907	93.3	0.913
Cross-entropy Loss + Topological Loss	93.6	0.916	94.8	0.932
Focal Loss	93.4	0.913	93.9	0.920
Focal Loss + Topological Loss	**94.2**	**0.923**	**95.9**	**0.947**

that combining Focal Loss and Topological Loss yields more accurate and topologically consistent wetland classification.

IV. DISCUSSION

In coastal wetland remote sensing classification, traditional deep learning models face significant challenges when distinguishing between land-cover types with similar spectral and textural features. These challenges arise not only from the limitations of visual cues, but also from the lack of domain-level understanding. The integration of KGs with deep learning addresses these issues by introducing structured prior knowledge to guide spatial representation learning.

First, KGs help compensate for the limitations of purely visual features through multimodal representation. Wetland vegetation categories often rely on semantic attributes and spatial patterns that are difficult to extract from pixel-level information alone. In our CRLNet, the CAKE module encodes graph-based attribute information into semantic vectors using BERT and fuses them with visual features. This enables the model to consider both visual appearance and semantic meaning. The STRE module further encodes spatial constraints using graph convolution based on topological relations. Together, these components construct a visual-semantic-spatial representation that significantly enhances fine-grained classification performance.

Second, KGs act as a form of regularization to alleviate overfitting. In noisy environments, models can easily be misled by local textures. The STRE module addresses this by applying a geometric loss that reduces the feature distance between adjacent categories (e.g., sea and seagrass beds), while increasing the distance between unrelated categories (e.g., tamarisk and sea). Meanwhile, the CAKE module leverages a gating mechanism to embed attribute knowledge—for example, emphasizing that *Phragmites australis* (common reed) generally has a greater plant height than *S. alterniflora*. This allows the model to focus on essential differences between categories, improving its generalization ability.

Finally, the use of KGs helps ensure that the model's predictions align with ecological and geographic logics. STRE promotes feature aggregation for co-occurring categories (e.g., *Spartina* and sea) via graph convolutions, avoiding topological contradictions. At the same time, CAKE's cross-modal attention aligns visual features with attribute embeddings, ensuring that predictions are both statistically and botanically coherent.

V. CONCLUSION

In this article, we propose a knowledge-driven approach for interpreting remote sensing images of coastal wetlands, leveraging category representation learning to enhance land-cover classification. By incorporating spatial topological relationships and attribute knowledge through graph convolution and cross-attention mechanisms, our method significantly improves classification accuracy, facilitating a more precise interpretation of coastal wetland land covers. A notable strength of CRLNet lies in its lightweight architecture, containing only 1/55th of the parameters of CGGLNet, thus ensuring computational efficiency without compromising performance. The experimental results on the Huanghe River Estuary and Yancheng coastal wetland datasets validate that CRLNet not only achieves superior accuracy but also operates with reduced complexity, making it particularly well-suited for deployment in resource-constrained environments.

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